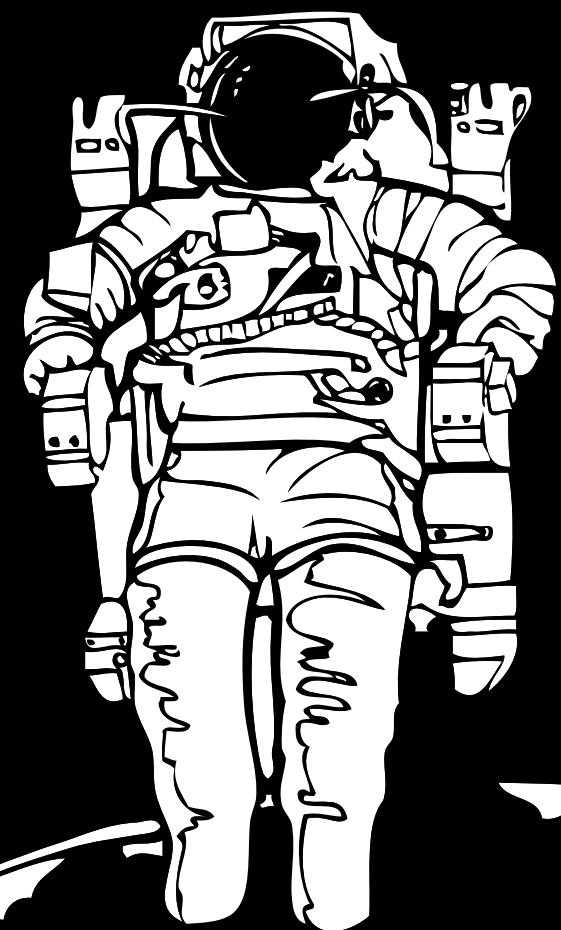


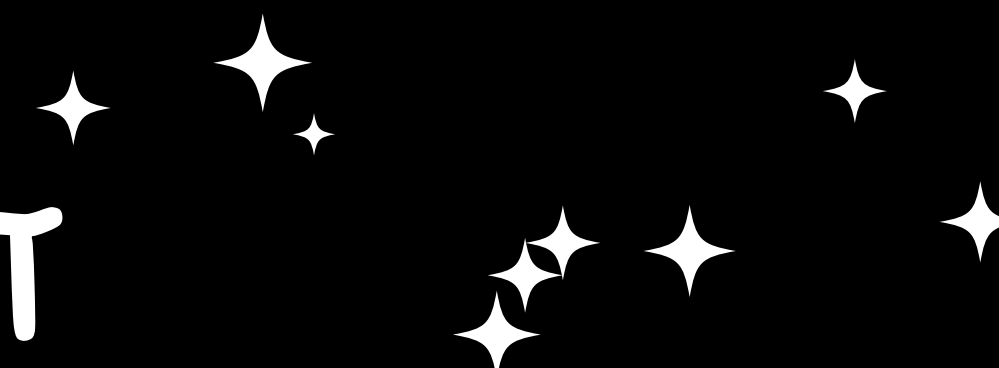


DSA IDEATHON

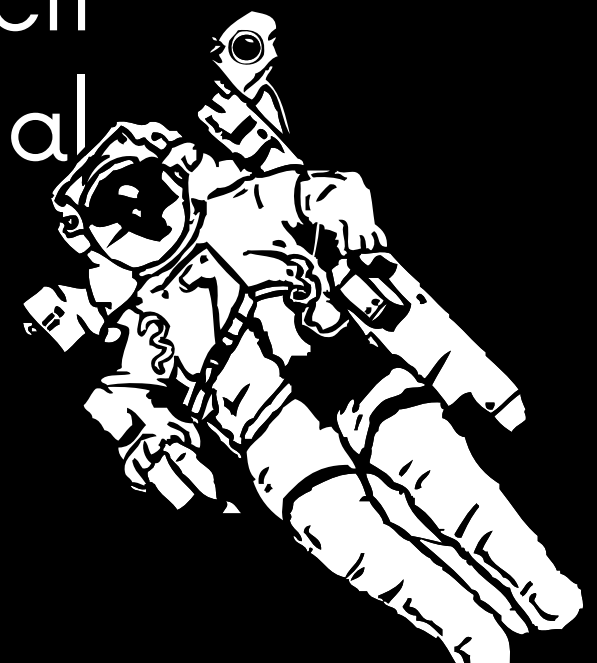


A diagram showing several concentric elliptical orbits around a central point, with small circles representing satellites or objects in orbit.

PROBLEM-STATEMENT

A cluster of stylized four-pointed stars of varying sizes in the top right corner.

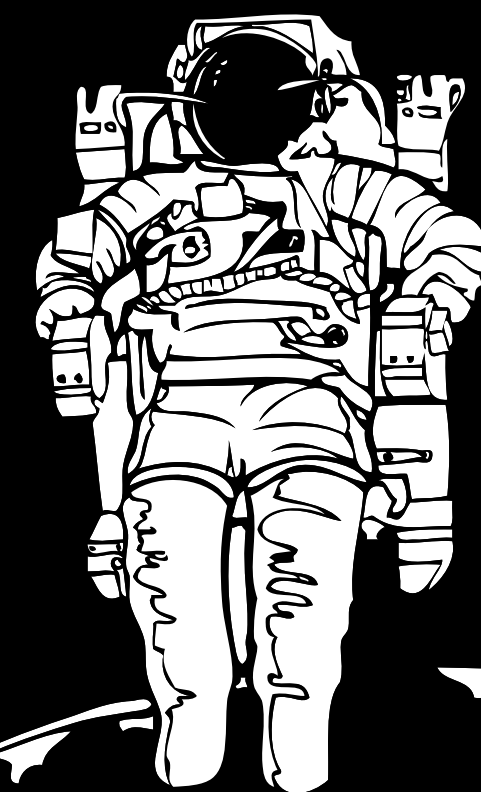
The increasing amount of space debris in Earth's orbit has made satellite collisions a serious risk. Even small fragments can cause major damage due to high orbital speeds. Operators need a fast and reliable way to predict potential collisions and plan safe maneuvers. This project builds a system that detects possible collision paths and suggests optimal avoidance routes using heuristic search algorithms, helping satellites navigate safely with minimal fuel use.





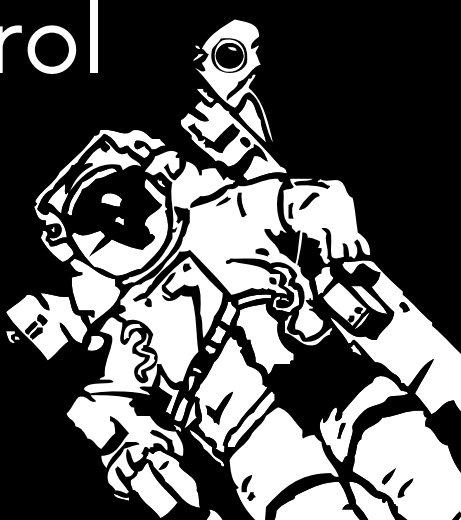
OBJECTIVE

- Predict potential satellite–debris collisions in advance.
- Generate safe and efficient avoidance paths.
- Minimize fuel usage and mission disruption.
- Provide a fast, automated decision–support system for satellite operators.



A diagram in the top left corner shows a central point with several concentric elliptical orbits. Two small circles representing satellites are positioned on different orbits. The top right corner features a cluster of white, four-pointed star shapes of varying sizes.

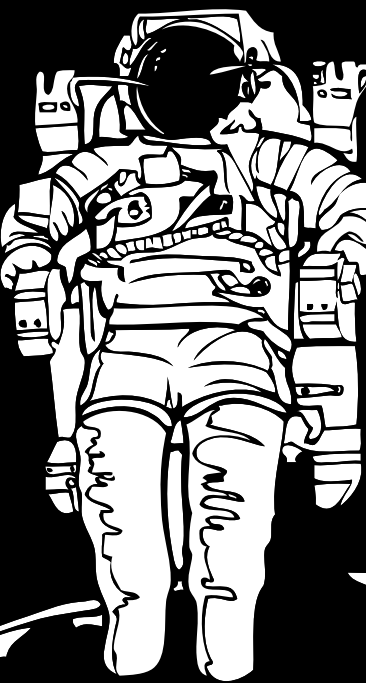
SOLUTION OVERVIEW

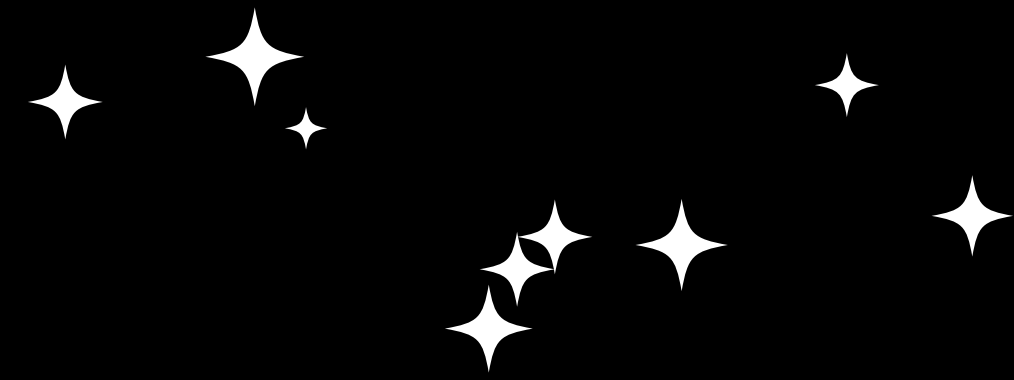
- Model satellites and debris using positional data.
 - Detect collision risks based on trajectory analysis.
 - Use heuristic search algorithms (like A^*) to find optimal avoidance routes.
 - Suggest maneuver solutions that keep satellites safe while conserving fuel.
 - Provide outputs that can be integrated into mission control workflows.
- 
- A stylized white line drawing of an astronaut in a full space suit, floating in the bottom right corner of the slide.



CONCLUSION

- The system successfully identifies potential satellite–debris collisions before they occur.
- Heuristic pathfinding ensures safe and fuel-efficient avoidance maneuvers.
- Automating the process reduces human error and speeds up decision-making.
- This approach improves satellite safety and contributes to long-term sustainability in Earth's orbit.





THANK YOU

Nimmagadda Princy
Jaikritha
Shreya Sardar
Uthkarsha

